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Top Skills

Data-driven Decision Making

Market Pricing

Marketing Strategy

Languages

English (Native or Bilingual)

Chinese (Native or Bilingual)

Peter Chang

Senior Director, Head of Marketplace Engineering at Coupang
Bellevue, Washington, United States

Summary

Build the Coupang Marketplace to provide sellers the most effective growth solutions in the world.

Experience

Coupang

9 years 8 months

Senior Director, Head of Marketplace Engineering

October 2022 - Present (3 years 9 months)

Greater Seattle Area

I am currently leading a global organization of talents in US, KR, and CN to evolve our marketplace service and deliver double-digit revenue growth year-over-year. Beyond its basic functions such as listing, inventory management, order management, reporting, marketing, and settlement, we are adding innovative features with cutting-edge technology to our sellers and customers. Here are some interesting and challenging topics we are working on.

- Product quality improvement, such as autofill metadata, picture matching & upgrading, description enrichment, etc.
- Keep prices competitive throughout the e-commerce marketplace.
- A self-service marketing tool for automatic promotion.
- Recommend products, prices, and opportunities to sellers in order to optimize selection.
- Enhancements to security and privacy for customers.

I prefer a flat organization structure (3 layers) in order to encourage collaboration and joint effort. Marketplace Engineering's major business units are:

- Marketplace Platform: responsible for developing existing and evolving features, as well as maintaining high reliability and security.
- Marketplace Growth: responsible for customer experience, GMV, revenue, and conversion growth.

- Other Bets: global sellers operation and growth, Coupang travel business, etc.

Director / Senior Director, Ads Infrastructure
February 2020 - October 2022 (2 years 9 months)
Greater Seattle Area

I moved to the Ads team in February 2020 and started building Coupang's Ads engine from scratch. I grew the team from 10+ engineers to 50+ engineers in 3 years. Our team achieved a billion-dollar revenue working with our talent business team in 2022. My business and technical contributions during this time earned me a promotion to the executive team. In general, the mission of my organization is to serve Performance Ads and Brand Ads solutions on Coupang. Specifically, its structure can be explained as follows:

- Ads Gateway: Provide internal and external interfaces for feeding ads.
- Ads Ranking: The core ML ranking engine serving ads from index, pre-ranking, ranking, filter, etc.
- Ads Creatives: Enhance creatives and UX flow on Coupang iOS, Android, and PC.
- Ads Budgeting: Manage and serve ads budgets in real-time.
- Ads Measurement: Provide reporting features and settlement to advertisers. In the meantime, it also generates ML features for the ranking engine.
- Ads Integrity: Provide fraud detection solutions while serving ads.

Director, Services Platform
September 2018 - February 2020 (1 year 6 months)
Seattle, Washington, United States

In 2018, I expanded the Data Services team to a global Service Platform team (40+ engineers in Seattle, Shanghai, and Seoul). Our team's job is to provide infrastructure services, tools, and frameworks for all domain customers to increase productivity and throughput. In detail, the service platform includes 4 sub-teams:

Messaging Platform

- Internal Messaging (PB per day): Managed Kafka Service, Canal Service (MySQL to Kafka), Roller Service (In-house product: Kafka to S3/ElasticSearch/HDFS)
- External Messaging (Millions of messages per day): Customer Communication Service for both transactional and marketing notifications including SMS, Email, iOS Push, Android Push, KaKao, and Line.

Weblogging Platform

- A centralized logging platform (1k data types and trillion events) to track and analyze customers' journeys in Coupang. It provides both a Hive interface for business analysts/data scientists to build data insights. It also provides a real-time event interface for ML feature generation in the search and discovery domain.

Common Frameworks and Tools

- This team provides a centralized deployment system, AWS resources and cost management system, A global API Gateway (Internal & Public)

Common Data Service

- Provided an aggregated catalog view (180+ columns) using Cassandra, Redis and Spark to merge multiple data sources.

Senior Manager, Data Services

March 2017 - September 2018 (1 year 7 months)

Seattle, Washington, United States

During this period, I transitioned from engineer to manager and built the data services team from scratch. Starting with the success of the Messaging Service team on its company-wide Kafka service, I developed two additional business values. One is a general Weblogging product, and another is a real-time data pipeline solution.

The Weblogging product "Lumberjack" is a frontend logging tool to track and analyze customers' journeys in Coupang. It includes mobile libraries (iOS, Android, web) and a backend data aggregation engine. Domain teams can easily use Lumberjack to extract and aggregate the business metrics they need, such as product impressions, clicks, orders, and related CTR, conversion, etc.

A real-time data pipeline solution was standardized after several products we developed based on Kafka and Spark. I led the team to consolidate some common data processing patterns and deployment tools together to launch a Coupang in-house real-time data pipeline to help domain teams increase productivity. Specifically, if a customer needs to run a data processing job, they only need to clone our Github repository. They can implement the interface, fill in required AWS metadata, then a job can be deployed in less than one hour.

My efforts have resulted in the growth of the data services team from 4 to 20+ including 2 line-managers in one and a half years.

Staff Software Engineer, Messaging Services

November 2016 - March 2017 (5 months)

Seattle, Washington, United States

Initially, I worked at Coupang Messaging Services as a staff engineer within the Platform team. In only five months, I led a small team of four engineers in launching the company-wide Kafka Service to replace the legacy RabbitMQ event distribution service. One of my most impressive accomplishments is bringing Kafka to front-end transactional services from the data analysis world, because not many engineers believed Kafka was stable enough to support P0 services in 2016. Fortunately, I convinced leadership to onboard the first and most critical customer, fulfillment & transportation service (FTS), which consumes and processes orders. As of today, it still functions efficiently and smoothly to support Coupang's growth.

Expedia, Inc.

Tech Lead

June 2015 - November 2016 (1 year 6 months)

Bellevue, WA

At Expedia Group, I played the tech lead role for the Transaction Bridge (TxB) team. TxB distributes and polishes customer orders from all upstream Expedia brands (Expedia, Hotels.com, Orbitz, etc.) to all downstream systems.

I led the team to build an end-to-end order distribution system for consumers who use orders to support their business. For example, the finance team consumes orders for settlement, the trip team consumes orders for trip planning, the data team consumes orders for business analysis and feature generation, the fraud detection team consumes orders for modeling, and so on. During my tenure, my team and I spent 12 months developing the product and 6 months selling it to 10+ domain teams.

The TxB product itself was built on Apache Kafka using Java, Apache Storm, Cassandra, Elasticsearch, Hive and Kibana. Kafka is the backbone of data flow. Storm is the computing engine and Cassandra is the main database. We utilize Elasticsearch to index metadata for real-time searching and reporting, while Hive is used for offline data analysis. In addition, the whole TxB is deployed on AWS with EC2 and Docker.

Visa

Senior Software Engineer

June 2014 - June 2015 (1 year 1 month)

Bellevue, WA

I worked at Visa for one year. And I was responsible for and contributed to the Proxy Connection Service (PCS)

PCS is a Java-based RESTful API between merchants and banks to validate and normalize payment requests such as Authorize, Capture, Settlement, Refund, etc. I worked with a few engineers to build this lightweight validation and logging layer from scratch in 6 months. In addition to delivering the product on time, I also achieved 95% code coverage. Furthermore, PCS successfully passed Fortify's scan, Veracode scan, IBM's App scan, and a 3rd party penetration test.

SAP Fieldglass

Software Engineer

June 2011 - June 2014 (3 years 1 month)

Naperville, IL

In my role as a full-stack software engineer at Fieldglass, I was responsible for both back-end and front-end projects and web services, which utilize Java, MySQL, Javascript, and other popular open source tools and frameworks. My major deliveries include:

- A messaging platform used to coordinate communications between employers, vendors and contractors.
- Represented Fieldglass in building ERP solutions for Chevron, Vodafone, and Bank of America using SOAP, REST and SFTP integrations.
- Perform SQL query tuning and data transmission optimization for customers' monthly, annual and ad-hoc reports to improve performance and reduce runtime.

Education

Illinois Institute of Technology

Master of Science (MS), Computer Engineering

Queen Mary University of London

Bachelor of Science (BS), Telecommunication Engineering with Management

Beijing University of Posts and Telecommunications

Bachelor of Science (BS), Telecommunication Engineering with Management